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Strategic drift in 10-K filings and S&P 500 returns

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Abstract

We study whether semantic drift in 10-K filings predict future stock returns among S&P 500 firms between 2005 and 2024. Using dense sentence embeddings generated by the all-MiniLM-L6-v2 transformer model, we compute a Semantic Drift Score for each firm-year separately across three sections of the 10-K: Item 1 (Business), Item 1A (Risk Factors), and Item 7 (MD&A). This approach captures shifts in meaning rather than surface-level word overlap and thereby extends the "lazy prices" framework of Cohen, Malloy and Nguyen (2020) beyond bag-of-words similarity to substantive repositioning in narrative content.

We have three main findings. First, drift in the Risk Factors section (Item 1A) is the dominant predictor of future returns: a long-short portfolio sorted on Item 1A drift earns a Fama-French five-factor alpha of -36.8 basis points per month (t = -2.92), whereas drift in Items 1 and 7 has weaker impact. Second, the predictive power is located at the twelve-month horizon and is insignificant at one to six month windows. Third, Fama-MacBeth interaction analyses show that the drift penalty is more severe for value firms and for smaller S&P 500 constituents, aligning with attention- and information-asymmetry explanations for delayed price discovery.

Methodologically, we show that transformer-based embedding distance outperforms TF-IDF similarity by capturing semantic repositioning even when firms rewrite disclosures using new vocabulary. Empirically, we establish that the informational content of 10-K filings is concentrated in the Risk Factors section. This is in contrast with prior work that aggregates textual change across the full document.

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1 Introduction

Annual 10-K reports filed to the U.S. Securities and Exchange Commission (SEC) are widely considered to be among the most informative disclosures of public listed firms. While traditional financial analysis focuses primarily on quantitative and accounting metrics such as earnings and cash flows, these filings have been proven to contain extensive narrative sections that provide meaningful information about management's perspective on strategy, applied or potential risks, forward guidance and performance. In particular, Item 1 (Business), Item 1A (Risk Factors) and Item 7 (Management's Discussion and Analysis of Financial Condition and Results of Operations MD&A) provide structured textual disclosures across firms, making them comparable and suitable for large-scale empirical analysis.

Our research is inspired by the "Lazy Prices" paper of Cohen, Malloy and Nguyen (2020) and by Hoberg and Phillips (2016). We construct a panel of 500 S&P 500 firms over January 2005 to December 2024, yielding 239 monthly cross-sections and approximately 112,035 firm-month observations. Firms enter the sample only during months in which they are constituents of the S&P 500 index and have a 10-K filing within the preceding twelve months. Firm identities are linked across databases using WRDS identifier mappings (CIK, GVKEY and PERMNO) and filings are retrieved from EDGAR and analyzed into sections. In order to ensure valid inference, the sample is restricted to yearly observations in which the firm is included in the S&P 500 and has an available 10-K filing, trying to tackle survivorship and look-ahead bias arising from previous disclosures and ensuring that only information available at the time is used.

The empirical framework focuses on measuring strategic shifts through changes in the semantic content of section 1, 1A and 7 of 10-K filings individually and compounded. These sections are strategically chosen because they provide executives and managers the greatest flexibility to communicate future strategic choices and firms' risk and thus reveal substantive shifts in strategic firm direction. While previous studies that focused on the industry strategic position, relied on dictionary-based methods like Bag-of-Words or TF-IDF cosine similarity which often fail to capture the semantic context of the documents, this study employs dense document embeddings generated by the all-MiniLM-L6-v2 transformer model. For each firm-year, Semantic Drift Scores are computed for each of the aforementioned sections separately and compounded as the complement of the cosine similarity between the section's yearly average of document embeddings in two consecutive years, producing values ranging from zero to one, with higher values indicating more intense strategic shift.

The firm-year panel is merged with stock returns and accounting variables from CRSP and Compustat. The predictive ability of the strategic shift textual signal is evaluated

using two complementary approaches. Initially, portfolio sorts examine whether sorting firms into quintiles based on each signal produces differences in future returns. Second, Fama-MacBeth regressions across sections test whether the signals maintain their explanatory power after controlling for standard firm characteristics such as size, profitability and momentum.

Our study contributes to the literature in several ways. First, we provide evidence on whether strategic shifts, captured through changes in the semantic content of Items 1, 1A and 7 of 10-K filings, are reflected in subsequent stock returns. By computing Semantic Drift separately for each of those sections and by comparing their predictive power, we identify that Section 1A provides the strongest signal for predicting subsequent returns, with Section 1 and 7 carrying a considerably weaker signal. Methodologically, our study contributes to the literature by proposing an innovative alternative way of measuring strategic change at the firm level: applying a neural network-based sentence transformer model (`all-MiniLM-L6-v2`) to Items 1A and 7 of annual 10-K filings. Traditional industry classifications, such as SIC codes provided by Compustat, rarely change even when a firm's underlying business has substantially evolved, which makes them poorly suited to capture strategic repositioning. In contrast, the Semantic Drift Score proposed in this study is capable of detecting strategic shifts directly from the language managers use to describe their business and risks, rather than from static industry labels.

Similarly to Cohen et al. (2020), our study is guided by the hypothesis that firms that exhibit higher strategic shift, as revealed by larger semantic shifts in Items 1, 1A and 7 of their 10-K filings, earn a lower future 12 month return than firms whose disclosures remain largely unchanged.

Combining the above, this framework provides an approach in evaluating the informational content of corporate disclosures. Combining a neural network-based measure of semantic drift with a carefully constructed, bias-controlled sample, the analysis aims to assess whether and how strategic shifts in narrative reporting are incorporated into asset prices.

2 Background and Related Literature

The application of Natural Language Processing (NLP) to 10-K corporate filings has expanded rapidly in recent years as computational power and the availability of textual data have increased

2.1 Textual data in financial economics

The use of textual data in financial economics stems from the idea that corporate disclosures contain valuable qualitative information that is not captured by accounting variables alone. Early evidence supports this view. F. Li (2008) shows that less readable annual reports are associated with lower future earnings, consistent with the notion that managers may obscure poor performance through complex language. Similarly, Tetlock (2007) and Tetlock, Saar-Tsechansky and Macskassy (2008) demonstrate that negative language in financial text can predict both earnings and stock returns, establishing that textual signals are informative, yet not fully incorporated into prices.

2.2 Bag-of-Words Approach (BoW)

The most widely adopted NLP technique in financial text analysis is the simple bag of words (BoW), which is commonly used for assessing sentiment. The BoW approach uses algorithms to simply categorize words based on different criteria by measuring the frequency of words or phrases, without considering the syntactic structure or the context of the sentences in which the words appear.

Several papers (Henry & Leone, 2016; F. Li, 2010) suggest that dictionaries that are built for general purposes might not be suitable for sentiment analysis of corporate disclosures, without undermining the results of previous studies. A key methodological development in this literature is the introduction of specific domain sentiment measures. Loughran and McDonald (2011) show that general purpose dictionaries tend to misclassify financial language in a systematic base and propose a dictionary specifically focused in finance, which has since become the standard for measuring tone in corporate disclosures. Loughran and McDonald (2011) developed a static, domain specific dictionary which categorized words into negative, positive, uncertainty, litigious, strong modal and weak modal and since then became the dominant dictionary in the field. Additionally, they demonstrated that approximately 75% of Harvard dictionary's negative words like tax or liabilities are misclassified words, that cannot be classified as negative in financial disclosures. This work represents a sentiment analysis in 10-K filings based on a dictionary and highlights the importance of tailoring linguistic tools to the institutional context needs.

Following the study of Loughran and McDonald (2011), the literature has expanded to thematic dictionaries. More specifically, Baier, Berninger and Kiesel (2020) created a ESG (Environmental, Social, Governance) dictionary while K. Li, Mai, Shen and Yan (2021) constructed a dictionary to measure corporate culture. The K. Li et al. (2021) dictionary is innovative as they use a semi-supervised machine learning approach and word embedding model (word2vec) to create the "culture dictionary".

2.3 Management Discussion and Analysis (MD&A)

The majority of the NLP literature focuses on the Management Discussion and Analysis (MD&A) section of 10-K filings as this is the only section where the managers have the flexibility to disclose information strategically, making it a valuable source of their intentions and future plans of the companies (F. Li, 2010).

Brown and Tucker (2011), focusing on yearly over yearly modifications in MD&A sections, find that intensity of changes in MD&A section are commensurate with the economic changes experienced by a firm. Additionally, they find that the MD&A section and 10-K filings in general have become less informative over the last years as managers rely more on boilerplate disclosures. Similar were the findings of K. Li et al. (2021), regarding 10K usefulness, who demonstrated that due to increased AI and Machine Learning readership in recent years, firms have shaped their filings in order to minimize negative sentiment for the stock. On the other hand, Cohen et al. (2020) demonstrated that major changes in 10-K still hold information value, by constructing a portfolio with long position on firms with unchanged filings and short on firms with major changes in 10-K filings, achieving a monthly alpha of 188 basis points.

2.4 Document Similarity Measures

Another approach that is widely adopted in NLP is to examine the similarity between documents. The most popular similarity measure is cosine similarity which calculates the cosine of the angle between the vector representations of two documents (Senave, Jans & Srivastava, 2023), producing values between 0 and 1. Identical documents will return a value of 1 while completely dissimilar documents will return 0. In terms of finance, Hanley and Hoberg (2012) used cosine similarity to measure the similarity between IPO prospectuses in order to examine firms' disclosure strategies and exposure to litigation risk. Hoberg and Lewis (2017) examine whether fraudulent firms produce abnormal disclosures by computing cosine similarity between a firm's MD&A disclosures and the average disclosure of its non-fraudulent industry peers with similar size and age.

Hoberg and Phillips (2016) construct text-based industry classifications from 10-K product descriptions and show that these outperform traditional SIC codes in capturing

competitive relationships. Their findings suggest that shifts in disclosure language reflect changes in firm strategy and economic positioning, motivating the use of thematic text measures to track strategic dynamics over time.

Another important part of research focuses not on the level of disclosure, but on how it changes over time. Cohen et al. (2020) show that firms with highly similar consecutive 10-K filings tend to underperform, contrary to those who substantially revise their disclosures. Textual change, typically measured using TF-IDF cosine similarity, captures the extent to which new information is being introduced. This result is often interpreted through a limited-attention framework (Hirshleifer & Teoh, 2003), in which investors are slower to process complex qualitative information. As a result, measures of text change provide a simple but powerful proxy to disclosure information and form a central component of this project.

2.5 NLP Models: BERT and Topic Modeling

Recent advances in NLP have introduced models based on transformers, which capture meaning dependent on context in financial text. A fundamental innovation in the field of textual analysis is the BERT model, an unsupervised pre-trained model introduced by Devlin, Chang, Lee and Toutanova (2019). Unlike traditional BoW methods that treat words as independent units, ignoring word order and context, the BERT model generates contextual token representations capable of capturing the semantic meaning of whole sentences and paragraphs. A firm describing itself as "a software company facing intense competition" and "a competition-intense company facing software" produces identical BoW vectors despite being semantically distinct; contextual embeddings separate the two cases. For disclosure-change studies this distinction is material, because managers can rewrite a section entirely while preserving its meaning, or retain most of its vocabulary while shifting its substance, and BoW-based similarity measures cannot tell the two apart.

For document-level similarity tasks, however, raw BERT is poorly suited. Its standard pooling (the `[CLS]` token) was not optimized to produce semantically meaningful sentence vectors, and cosine similarity between raw BERT embeddings correlates weakly with human judgments of semantic similarity. Reimers and Gurevych (2019) address this limitation with Sentence-BERT (SBERT), which fine-tunes BERT-family models using a siamese network architecture and a contrastive learning objective on sentence-pair datasets (natural language inference and semantic textual similarity). The result is an embedding space in which cosine distance directly reflects semantic dissimilarity, which is precisely the property our Semantic Drift Score relies on. The specific model we deploy, all-MiniLM-L6-v2, is a distilled six-layer variant of this family trained on over one billion sentence pairs. It produces 384-dimensional embeddings and, despite being smaller than BERT-base (22M versus 110M parameters), achieves competitive performance on stand-

ard semantic textual similarity benchmarks. Its small size is a practical advantage for processing the large corpus of multi-page 10-K sections in our sample.

Domain-adapted alternatives also exist within the BERT family. The FinBERT model, developed by Araci (2019) and Huang, Wang and Yang (2023), is pre-trained on financial corpora and is widely used for sentiment classification and ESG topics, accounting for syntax, negation, and semantic context in ways dictionary methods cannot. Huang et al. (2023), analyzing the sentiment of 10-K filings, find that FinBERT achieves significantly higher accuracy than the Loughran-McDonald dictionary as well as NB, SVM, RF, CNN, and LSTM algorithms. For our specific task, however, the trade-off is less clear: FinBERT is optimized for short-sentence sentiment classification rather than document embedding, and applying it to multi-paragraph business descriptions risks conflating tonal signals with strategic content. These methods also introduce a trade-off between predictive power and interpretability.

Another neural network model in the BERT family is the BERTopic model, a neural topic model developed by Grootendorst (2022). While traditional topic models such as Latent Dirichlet Allocation (Blei, Ng & Jordan, 2003) and Non-Negative Matrix Factorization (Févotte & Idier, 2011) may fail to accurately capture the topic of a document because they ignore the context of words in a sentence, BERTopic overcomes these limitations by combining pre-trained transformer-based document embeddings with a class-based TF-IDF. Jehnen, Teuteberg and Lucka (2025), analyzing 10-K filings, demonstrate that BERTopic combined with FinTextSim embeddings significantly outperforms LDA and NMF in identifying economic themes. As stated by the model's developers (Grootendorst, 2022), however, BERTopic has the crucial limitation of capturing only one topic per input. For our purposes, this discrete-topic structure is ill-suited to measuring continuous drift between documents, which motivates our choice of a dense-embedding approach.

Several features of our architecture matter for the interpretation of our results. First, all-MiniLM-L6-v2 has a 512-token input limit, which requires chunking long 10-K sections and aggregating the resulting embeddings. We adopt mean-pooling across chunks, which preserves the central tendency of a section's content but can dilute highly localized shifts; alternative aggregation schemes such as attention-weighted or max-pooling constitute a natural robustness test. Second, the training objective optimizes for cosine similarity rather than Euclidean distance, which motivates our choice of a cosine-based drift measure. Third, the pre-training corpus is general web text rather than financial disclosures; this limits the model's ability to capture finance-specific semantics but also reduces the risk of overfitting to financial buzzwords or boilerplate. A subtle related concern is pre-training data leakage: since 10-K filings are publicly accessible online, we cannot fully rule out that some filing in our sample appeared in the training corpus. However, the contrastive training objective does not reward memorization of document-level content,

so the concern is less severe than for LLMs fine-tuned directly on financial text.

Finally, a growing literature applies large language models directly to asset pricing questions. Lopez-Lira and Tang (2023) use ChatGPT to extract return-predictive signals from news headlines, and Kim, Muhn and Nikolaev (2024) show LLMs can perform financial statement analysis at the level of human analysts. These approaches differ from ours in using LLMs as zero-shot reasoners over text rather than as embedding functions, but they point to an active research frontier adjacent to the methodology we employ.

2.6 Defining strategic shifts

A strategic shift happens when a company fundamentally changes its direction or market focus to stay competitive. While traditionally viewed as one-off events, modern research shows that boards of directors actually see these changes as a series of connected moves (Klarner, Yoshikawa & Hitt, 2023). Accurately measuring and recognizing these shifts is important because they can have a direct impact on business potential and thus firm value. Historically, standard financial metrics have done a poor job of capturing the real value of these complex changes. For example, old financial models and research often suggested that diversification hurt a company's stock price ("the investors can diversify themselves"). Today, advanced machine learning and natural language processing tools help identify the moment changes take place, such as diversification. By using algorithms to quickly analyse many yearly reports, the company's true strategic direction can be observed. This new approach has proven that expanding and diversifying may, in fact, increase a firm's value (Choi, Chung & Kim, 2021).

2.7 Symbolic management and the hype phenomenon

When managers face pressure to change course or want to boost stock price quickly, they may be tempted to turn to symbolic management and submit to the hype rather than actually commit to a strategic shift. Executives use carefully chosen language in their public disclosures to frame their strategic shifts, a process that allows them to appease different stakeholders while "decoupling" their stated goals from their real, underlying operations (Fiss & Zajac, 2006). In essence, companies can create a smokescreen by changing how they talk about their business without actually changing the business itself.

This behaviour often leads to a "hype" phenomenon, where adopting trendy buzzwords, hypes or even changing the company name (think .com bubble) is used to artificially inflate market interest. Interestingly, investors even seem to reward these purely cosmetic or concept-driven changes with short-term stock price increases, proving that markets can be quite easily swayed by corporate framing and broader sentiment rather than substantive financial reality (Hui & Tan, 2022). By manipulating the narrative in their filings,

managers can attempt to manage public perception and (temporarily) boost firm value without actually making meaningful strategic changes.

2.8 Measuring shifts (NLP)

To effectively capture the increasingly complex modern strategic pivot points, researchers are increasingly using sophisticated textual analysis techniques, instead of simply relying on static financial ratios. SEC 10-K filings, especially the business description (item 1) and the management discussion and analysis (item 7) offer a good description of the management's strategic narrative. By applying natural language processing to these texts, one can construct dynamic industry classifications and more interestingly, the changes in this classification over time (Hoberg & Phillips, 2016). This NLP-approach maps the evolution of corporate strategy better than traditional, rigid industry codes. Furthermore, analyzing the tone of these public filings provides powerful predictive signals regarding future operations. For example, using machine learning algorithms to evaluate the forward-looking statements within the MD&A section uncovers nuanced tonal shifts that significantly predict future firm performance and liquidity (F. Li, 2010). Ultimately this transition to NLP enables financial researchers to quantify the subtle, qualitative language that executives use to frame their corporate strategy and thus predict how investors actually process these disclosures.

2.9 Market pricing and the lazy prices effect

Despite the amount of information embedded in corporate filings, the stock market often fails to efficiently and accurately process textual changes. While investors may react quickly to surface-level hype and buzzwords, seemingly rewarding hollow strategic announcements with immediate, short-term stock price hikes (Hui & Tan, 2022), they seem to underreact to deeper substantive modifications deep inside official SEC filings. This behavioural anomaly is captured as the "lazy prices" phenomenon, which demonstrates that investors are generally not attentive to textual shifts within regular financial disclosures (Cohen et al., 2020). Specifically, when management alters the language in the risk factors (item 1A) or the MD&A (Item 7), these modifications serve as highly predictive signals of future changes in firm operations.

Because investors do not immediately price in these textual changes, firms that significantly alter their reporting language often experience predictable, long-term abnormal returns once the true underlying value in those changes reveal themselves (Cohen et al., 2020). Therefore, by tracking these specific reporting discrepancies, traders using machine learning models can attempt to exploit this market inefficiency and try to accurately forecast this long-term abnormal performance.

Overall, the past literature suggests that corporate disclosures contain multiple layers of information which are not fully reflected in market prices and can be captured using specific approaches. We test whether deep semantic representations improve asset pricing signals beyond traditional text similarity.

3 Data

3.1 10-K Forms

For the collection of the data, our objective is to extract 10-Ks from companies in the time that they were part of the S&P 500. For this, a Python script is deployed that extracts components of the 10-K that are needed for our study, rather than the entire 10-K. This is done to optimize our storage and speed of using this data in our research. We pull the full 10-K from U.S. Securities and Exchange Commission's EDGAR database (U.S. Securities and Exchange Commission, 2025) and match sections: Item 1, Item 1A and Item 7 through regular expressions. We pull the 10K's in parallel across different cores and store them in parquet files based on cusip in google drive. The content of these items and their significance in our study are as follows:

The item 1 section requires management to provide a detailed description of the company's operations, products, services and the markets in which it competes. It includes information regarding the firm's business model, competitive positioning and intellectual property. Item 1 serves as the primary Semantic Signal for the firm's Strategic Intent. Within the narrative of this section, management explicitly defines what the company does and how that focus is evolving. By applying our all-MiniLM-L6-v2 architecture to this item, we can quantify Strategic Semantic Drift. For example: A transition in the thematic clusters of Item 1 from Hardware Manufacturing to Artificial Intelligence represents a measurable strategic pivot.

In item 1A companies are mandated to disclose the most significant factors that make an investment in the company speculative or risky. These disclosures range from macroeconomic concerns and regulatory hurdles to specific operational risks. Item 1A represents the Linguistic Effect of a strategic shift. We use this section as our primary Dependent Variable for Risk. By applying our all-MiniLM-L6-v2 in this section, we test whether year-over-year shifts in the regulatory and operational risks faced by the firm provide a statistically significant textual signal for future returns.

Item 7 provides a narrative explanation of the firm's financial performance from the perspective of management. It focuses specifically on liquidity, capital resources and results of operations, often providing a bridge between the financial statements and the narrative strategy.

3.2 Financial Data

The financial data is extracted from two WRDS databases (Wharton Research Data Services, 2026) using the python API. Accounting variables, including total assets, net income, R&D, EBIT, Book to market and cash, are sourced from Compustat. Returns are sourced from CRSP and momentum and reversal are derived from there aswell.

The two databases are then linked at the firm-year level using the 8-digit CUSIP identifier, which is common to both Compustat and CRSP. Market value is calculated as the product of share price and shares outstanding. Momentum is computed with the compounded return of months t-12 to t-2. We skipped the most recent month to avoid short-term reversal effects. We use the log form of the market value to comply with Fama-French standards.

To ensure the drift signal uses only information publicly available at the time of prediction, we align returns to filing dates as follows. For each 10-K filed on date d, the firm's drift score is assigned to every monthly cross-section beginning on the first day of the calendar month strictly after d. For example, a 10-K filed on February 24 generates a drift score that enters the March cross-section and predicts the March return. The score remains assigned to that firm until the subsequent 10-K is filed, at which point it is replaced. This implies each annual drift observation informs, on average, twelve monthly cross-sections. This timing convention guarantees that no return month t uses textual information that was not publicly disclosed at $t-1$, eliminating look-ahead bias from the filing-date channel.

3.3 Control Variables and Sample Construction

We begin with all 10-K filings submitted by firms that were S&P 500 constituents at any point between January 2005 and December 2024, retrieved from EDGAR ($N \approx 10K$ raw filings across 868 unique CIKs). We parse Items 1, 1A, and 7 using regular expressions.. We link filings to CRSP returns via the CIK-to-PERMNO mapping from WRDS, requiring a match for monthly return data. We merge Compustat accounting data (total assets, net income, book equity) on GVKEY and require non-missing book-to-market and ROA at the fiscal-year end preceding the return month. Finally, we restrict each firm-month observation to periods in which the firm is an active S&P 500 constituent. The final estimation panel comprises 686 unique firms, 239 monthly cross-sections (January 2005 – December 2024), and 112,035 firm-month observations. Filing dates are taken from the EDGAR filing timestamp; a firm's drift score enters the cross-section in the first calendar month that begins strictly after its filing date, and remains assigned to that firm until the next 10-K is filed, typically twelve months later.

Building upon the standard asset pricing literature (Fama & French, 1993; Carhart,

Table 1: Summary Statistics and Correlations

Panel A: Summary Statistics

Variable	Obs	Mean	Std.	p10	Median	p90
Drift: Item 1	111023	0.024	0.023	0.007	0.018	0.047
Drift: Item 1A	99825	0.023	0.025	0.007	0.015	0.044
Drift: Item 7	111259	0.027	0.062	0.005	0.016	0.036
Drift: Composite	112240	0.026	0.033	0.009	0.018	0.042
Monthly Return	112035	0.010	0.084	-0.091	0.011	0.107
Momentum (12-2m)	111992	0.118	0.300	-0.239	0.104	0.475
Reversal (1m)	112188	0.010	0.078	-0.085	0.012	0.100
log(MktCap)	112041	9.816	1.144	8.427	9.702	11.432
Book-to-Market	108717	0.465	0.381	0.102	0.359	0.953
ROA	111887	0.065	0.067	0.006	0.057	0.151

Panel B: Correlation Matrix (Drift Signals)

	Item 1	Item 1A	Item 7	Composite
Item 1	1			
Item 1A	0.343	1		
Item 7	0.144	0.182	1	
Composite	0.526	0.634	0.841	1

Notes: Panel A reports summary statistics for the main variables used in the analysis. Drift variables are constructed from textual sections of 10-K filings. Returns are monthly. Panel B reports pairwise Pearson correlations between drift measures. The composite measure is the mean of the three drift items.

1997; Jegadeesh & Titman, 1993), we construct firm-level characteristics to control for size, value, momentum and short-term reversal effects. Firm size is controlled for using the natural logarithm of market capitalization (`log(MktCap)`), which accounts for the extreme positive skewness in raw equity values. The value premium is captured using the book-to-market ratio (`Book/Market`), formally defined as the book value of common equity divided by the market value of equity at the fiscal year-end. Additionally, to account for the well-documented tendency of past winners to continue outperforming, we include a momentum control (`Momentum`), calculated as the compounded stock return from month $t-12$ to $t-2$. Finally, we control for short-term microstructure frictions and liquidity effects by including a one-month lagged return reversal variable (`Reversal`), measured as the firm's return in month $t-1$.

To align with the more extended Fama and French (2015) five-factor framework, we expand our control suite to account for firm profitability, which is a well-documented driver of cross-sectional expected returns. Operating profitability is explicitly captured using Return on Assets (`ROA`), calculated as net income scaled by total assets. Controlling for this operational metric is crucial in our text-based setting, as firms undergoing major strategic shifts might simultaneously exhibit temporary profitability shocks that could otherwise confound the semantic drift signal. While the standard five-factor model also

incorporates an investment factor (typically asset growth), we omit it from our primary cross-sectional specification to maintain parsimony and avoid potential multicollinearity with our existing fundamental controls.

Ultimately, utilizing this comprehensive suite of controls provides a robustness check for our NLP signal. While the standard size, value, momentum and reversal variables address the most prominent cross-sectional market anomalies, the addition of the profitability metric accounts for the underlying financial mechanics that often accompany strategic repositioning. By controlling for these fundamental financial characteristics, we can confidently conclude whether the semantic drift observed in 10-K filings contains unique, unpriced forward-looking information or if it is merely a textual reflection of existing accounting phenomena.

Table 2: Descriptive Statistics: Fama-French Six-Factor Returns

Variable	**Obs**	**Mean**	**Std. Dev.**	**Min**	**Max**
Market Excess Return (MKT)	249	0.811	4.413	−4.886	5.602
Size Factor (SMB)	249	−0.007	2.664	−3.218	3.284
Value Factor (HML)	249	−0.079	3.163	−3.250	3.488
Profitability Factor (RMW)	249	0.357	1.866	−1.962	2.480
Investment Factor (CMA)	249	−0.014	1.885	−1.992	2.076
Momentum Factor (UMD)	249	0.108	4.370	−4.384	4.320
Risk-Free Rate (RF)	249	0.128	0.155	0.000	0.400

Notes: Monthly Fama-French six-factor returns in percent.

4 Methodology

4.1 Strategic Semantic Shifts via Sentence Embeddings

To detect year-over-year shifts in the deployed strategy of a firm, we apply the pre-trained *all-MiniLM-L6-v2* model from the Sentence-Transformers framework (Reimers & Gurevych, 2019; Sentence-Transformers, 2021) to Item 1,1A and 7 separately and compounded of each 10-K filing. This model generates dense 384-dimensional document embeddings that encode semantic meaning rather than surface-level word frequency. It is specifically trained and optimized for semantic textual similarity tasks using contrastive learning objectives, making it well-suited for comparing the strategic narrative of the same firm across consecutive years. Two filings that discuss the same strategic concepts, but with different vocabulary, will be mapped to nearby regions of the embedding space. Two filings that share the same vocabulary, but describe fundamentally different strategies will be mapped far apart. This model is therefore best suitable for our research.

Our choice of *all-MiniLM-L6-v2* over domain specific alternatives, like FinBERT,

stems from the objective of the model. FinBERT is used for short-sentence financial sentiment classification, which is a different task from document-level semantic similarity. Applying a model like FinBERT to long-form business descriptions would risk combining tonal signals with strategic content and could potentially introduce noise. *all-MiniLM-L6-v2* is optimized to preserve semantic similarity structure, so that we can directly interpret the cosine distances as measures of strategic repositioning.

Since the MiniLM-L6-V2 model has a maximum input limit of 512 tokens and the section 1A and Section 7 filings typically exceed this limit, each of the examined document section is split as described above into fixed size chunks and for each of those chunks k a 384-dimensional embedding is generated.

A concern with any pretrained embedding model in a asset pricing setting is data leakage. all-MiniLM-L6-v2 was trained on general web text crawled before our sample ends, and 10-K filings are publicly accessible online. We thus cannot rule out that some filings in our sample appeared in the training corpus.

The documents embedding is computed as the mean of all chunk embeddings as follows:

$$V_{i,t}^{k} = \frac{1}{K} \sum_{k=1}^{K} \text{embedding}(\text{Chunk}_k) \tag{1}$$

The semantic drift score for section k of firm i in year t is defined as:

$$\text{Drift}_{i,t}^{k} = 1 - \frac{V_{i,t}^{k} \cdot V_{i,t-1}^{k}}{\|V_{i,t}^{k}\| \cdot \|V_{i,t-1}^{k}\|} \tag{2}$$

where $V_{i,t}^{k}$ is the embedding of section k for firm i in year t. The Composite Semantic Drift Signal is calculated as the mean of the three individual Semantic Drift Signal Scores of Section 1,1A,7.

A value of 0 indicates that the firm's Item 1 embedding is identical across the two consecutive years, meaning that there is no strategic shift. A value of 1 or close to 1, means that the two consecutive years are on different regions of the embedding space. This can be interpreted that there is indeed a strategic shift between the years. With the value of 1 indicating a complete strategic repositioning.

Compared with the alternative simple word-based similarity measures that has been used by Hoberg and Phillips (2016), our approach focuses on the meaning of text rather than simply matching words, making it more effective in capturing semantic divergence in the 10K-S disclosures.

4.2 Portfolio Sorts

In order to evaluate whether the NLP strategic shift signal $SemanticDrift$ contains critical information about future stock returns, we employ a portfolio sorting approach following Cohen et al. (2020). The quintile assignment is updated once per year when a new 10-K is filed. Firms thus stay in their quintile for twelve months after the filing date. The Q_1 quintile contains the firms with the lowest signal values while Q_5 contains those with the highest. Each firm is then held in its assigned quintile portfolio for the twelve months following the fiscal year-end. Monthly portfolio returns are computed as both equal-weighted averages of the constituent stock returns, which are retrieved from CRSP-Compustat database.

The abnormal returns are computed with CAPM, the Fama and French (1993) three factor model,the extended version of FF3 with a momentum factor (MOM) introduced by Carhart (1997) and the Fama and French (2015) five factor model.

The market factor controls for aggregate market risk, SMB captures return differences related to size between small and large firms and HML accounts for the value premium associated with high compared with low book-to-market firms. With the addition of momentum, we try to control for the possibility of past winners continue to outperform and past losers to underperform. RMW and CMA account for profitability and investment patterns,

Capital Asset Pricing Model (CAPM). (Sharpe, 1964)

$$R_t^{LS} - R_t^f = \alpha + \beta_{\text{MKT}}\,(\text{MKT} - R^f)_t + \varepsilon_t \tag{3}$$

Fama-French Three-Factor Model (FF3). (Fama & French, 1993)

$$R_t^{LS} - R_t^f = \alpha + \beta_{\text{MKT}}\,(\text{MKT} - R^f)_t + \beta_{\text{SMB}}\,\text{SMB}_t + \beta_{\text{HML}}\,\text{HML}_t + \varepsilon_t \tag{4}$$

Carhart Four-Factor Model (FF3 + Momentum). (Carhart, 1997)

$$R_t^{LS} - R_t^f = \alpha + \beta_{\text{MKT}}\,(\text{MKT} - R^f)_t + \beta_{\text{SMB}}\,\text{SMB}_t + \beta_{\text{HML}}\,\text{HML}_t + \beta_{\text{MOM}}\,\text{MOM}_t + \varepsilon_t \tag{5}$$

Fama-French Five-Factor Model (FF5). (Fama & French, 2015)

$$R_t^{LS} - R_t^f = \alpha + \beta_{\text{MKT}}\,(\text{MKT} - R^f)_t + \beta_{\text{SMB}}\,\text{SMB}_t + \beta_{\text{HML}}\,\text{HML}_t + \beta_{\text{RMW}}\,\text{RMW}_t + \beta_{\text{CMA}}\,\text{CMA}_t + \varepsilon_t \tag{6}$$

All factor data are obtained from Kenneth French's data library. The intercept α in each specification is the risk-adjusted abnormal return and represents the component

of the long-short spread unexplained by any of the included systematic factors. Statistical inference is based on Newey-West heteroskedasticity and autocorrelation consistent standard errors.

4.3 Cross-Sectional Regressions

The central question to be answered is whether firms that have experienced a strategic shift between the years exhibit differences in risk-adjusted returns in the year following their 10-K filing. While the portfolio sorts provide an intuitive test regarding the relation between $SemanticDrift$ and future returns, they do not control for specific firm characteristics known to predict the cross-section of returns. The same question is also applied to the broader set of text-change signals, where Cohen et al. (2020) predict that larger disclosure changes signal negative forward-looking information that is not yet reflected in prices.

In order to examine if $SemanticDrift$ predicts future company's returns in the cross section, we implement the Fama and MacBeth (1973) regression procedure. For each month t, we estimate the cross-sectional regression of firms' returns, based Drift Signal and a set of firm-specific control variables:

$$R_{i,t+1} = \beta_{0,t} + \beta_{1,t} \cdot \text{SemanticDrift}_{i,t} + \sum_k \beta_{k,t} \cdot \text{Controls}_{i,t} + \varepsilon_{i,t+1} \tag{7}$$

where $R_{i,t+1}$ is the monthly return of firm i in month $t+1$, R_t^f is the risk-free rate and $\text{SemanticDrift}_{i,r(t)}$ is the drift score measured from firm i's 10-K filing, as described above. The signal is derived as soon as a new 10-K is published and is used to predict the returns in the following one, three, six and twelve months. The one-month lag ensure that only information available to the public at t is used to predict returns at time $t+1$. In that way, look-ahead bias is avoided, ensuring that no forward looking information enters the regression, avoiding look-ahead bias.

The vector $Controls_{i,t}$ includes the standard firm characteristics used and risk exposures. More specifically it includes log(MktCap), Book-to-Market, ROA, Momentum and Short-term Reversal. This ensures we isolate the incremental predictive content of the drift signal, we adopt the standard dual approach in empirical asset pricing. First, firm-level characteristics such as size, book-to-market, momentum and profitability are included as controls in the cross-sectional regressions, capturing well-documented return predictors. Second, in complementary time-series regressions, we evaluate long-short portfolio returns relative to standard asset pricing factor models (Fama-French and Carhart), which proxy for systematic risk exposures. The distinction between risk-based and characteristic-based explanations is not well defined in asset pricing literature, because we use both this allows

us to assess whether the drift signal contains predictive information in both cases.

The coefficient of primary interest is β, which measures the marginal predictive power of the signal for future excess returns after controlling for firm characteristics in the cross-section. It captures whether firms with larger semantic shifts in their disclosures earn different subsequent returns. A negative and significant β, consistent with the finding of Cohen et al. (2020), would indicate that firms whose disclosures change more earn lower future returns.

In the second step, the estimated coefficients β_t are averaged over time to obtain the overall coefficient:

$$\bar{\beta} = \frac{1}{T} \sum_{t=1}^{T} \beta_t \tag{8}$$

This average coefficient indicates the expected cross-sectional relationship between the signal and future returns. Statistical inference is based on time-series standard errors with Newey and West (1987) adjustments to account for potential heteroskedasticity and autocorrelation.

A high value in SemanticDrift indicates that there is a difference in the way the company describes itself from the past year or its strategy, operations or outlook has shifted in a important grade. The above may reflect an entry into new markets, changes in business model, restructuring operations or even a reaction to a shock. In short, it indicates a strategic instability or a repositioning. From an economic perspective, such changes may signal an increased uncertainty about their outcome. So, when a negative relation between semantic drift and subsequent returns are observed, indicates that investors might be skeptical about placing their bets on the company. The increased risk of this transition can make the returns lag, until positive results will be demonstrated. On the other hand, a positive relation would indicate that a strategic shifts created changes that increased company's value,and was recognized by the market at the upcoming period. Concluding, the regression tests whether such textual changes regarding the strategy shift are priced by the market.

4.4 Horizon Analysis

To evaluate the speed at which the market processes and prices the strategic shifts captured by our NLP signals, we extend our baseline asset pricing tests across multiple holding-period horizons. Under the strong form of the efficient market hypothesis, value-relevant information embedded in 10-K filings should be priced almost immediately upon publication. However, the "lazy prices" framework posits that investors suffer from limited attention and bounded rationality, causing a delayed market reaction to complex,

qualitative textual modifications.

To empirically test the speed of information incorporation, we vary the return prediction window. We evaluate the long-short portfolio spreads and the Fama-MacBeth cross-sectional coefficients over $H = 1, 3, 6$ and 12 month horizons following the filing date. If the market is systematically slow to process semantic drift, we hypothesize that the predictive power of the signal will be statistically insignificant at short horizons (one to six months) and will only become statistically significant at the twelve-month horizon, as the underlying strategic pivots or risks eventually materialize in the firm's fundamental performance.

4.5 Interaction Effects

To further isolate the economic channels driving the return predictability of semantic drift, we introduce interaction terms into our Fama-MacBeth cross-sectional framework. If the negative return predictability is truly driven by investor inattention and information asymmetry, the effect should not be uniform across all types of firms; rather, it should be more pronounced in specific cross-sections of the market.

To test this, we augment our baseline regression by interacting the lagged semantic drift score with key firm characteristics, specifically firm size (log(MktCap)) and the value premium (Book/Market). We hypothesize a positive interaction coefficient with firm size, which would indicate that the negative drift signal is stronger (more negative) among smaller firms. Smaller firms typically suffer from lower analyst coverage, lower institutional ownership and less algorithmic machine readership, allowing negative textual signals to remain unpriced for longer periods. Similarly, we interact the drift signal with the book-to-market ratio to test for value-versus-growth effects. Value firms are typically mature businesses; a sudden, radical shift in their strategic narrative or risk profile often implies severe operational distress or high execution risk. Consequently, we expect a negative interaction coefficient here, indicating that value firms are penalized more heavily by the market for high semantic instability than their growth-oriented counterparts.

5 Results

This section of our paper examines the results of four empirical tests conducted to evaluate whether higher semantic drift in Sections 1, 1A and 7 of 10-K filings predicts future stock returns. We first examine the unconditional predictive power of each signal through univariate quintile portfolio sorts. We then test whether the signals survive risk adjustment under a nested sequence of factor models: the CAPM, the Fama-French three-factor model, the Carhart four-factor model and the Fama-French five-factor model. Third, we estimate Fama-MacBeth cross-sectional regressions that control for standard firm char-

acteristics known to predict the cross-section of returns. Finally, we assess the economic mechanism underlying our results through horizon decomposition and interaction analyses. Throughout, statistical inference is based on Newey-West heteroskedasticity- and autocorrelation-consistent standard errors with six lags.

5.1 Portfolio sorts

Table 3 reports equal-weighted average monthly returns for portfolios sorted on each of the four semantic drift measures. For each signal, firms are assigned to quintiles based on drift measured within the filing year and equal-weighted portfolios are held over the twelve months beginning the month after the 10-K filing date.

The Q1 low-drift portfolios earn economically large monthly returns of between 0.89% (Item 1) and 1.07% (Item 1A), all statistically significant at conventional thresholds. In contrast, Q5 high-drift portfolios earn systematically lower returns, with the differential most pronounced for Item 1A (Q1 = 1.07%, Q5 = 0.81%). The resulting long-short spread for Item 1A of −0.247% per month ($t = -2.11$) or approximately 2.92% annualized, is statistically significant at the 5% level. Critically, this pattern is consistent with the finding of Cohen et al. (2020) that greater disclosure change predicts lower future returns, although we measure change via semantic embedding distance rather than TF-IDF word overlap. The long-short spreads for Item 1, Item 7 and the composite measure are economically small and statistically indistinguishable from zero (t-statistics of −0.17, −0.19 and −0.38 respectively). This heterogeneity across sections is itself an important finding. It indicates that the disclosure section most tightly regulated by materiality considerations the risk factors mandated by Regulation S-K carries the strongest predictive content, whereas unstructured narrative sections such as the business description and the MD&A convey no unconditional signal. One interpretation is that managers enjoy substantial latitude in how they describe the firm's operations and its financial performance and this latitude manifests as noise in the embedding space. Item 1A, by contrast, requires managers to enumerate material risks whose inclusion or exclusion has direct legal consequences, so changes in the language of this section plausibly reflect genuine revisions in the manager's perceived risk landscape. The fact that the composite signal, which averages across all three sections, displays no significant spread is consistent with this interpretation: averaging the informative Item 1A drift with the noisy Item 1 and Item 7 drifts dilutes the overall signal.

Table 3: Quintile Portfolio Sorts

Signal	Portfolio	Mean (%/month)	Ann. Return (%/year)	Volatility (%/year)
Item 1	Q1	0.890***	11.215	15.792
		(3.02)		
	Q5	0.875***	11.021	16.392
		(2.86)		
	Q5-Q1	−0.015	−0.176	4.710
		(-0.17)		
Item 1A	Q1	1.072***	13.644	16.619
		(3.39)		
	Q5	0.806***	10.109	16.579
		(2.60)		
	Q5-Q1	−0.247**	−2.918	6.124
		(-2.11)		
Item 7	Q1	0.916***	11.560	15.595
		(3.14)		
	Q5	0.895***	11.284	15.959
		(3.00)		
	Q5-Q1	−0.021	−0.250	5.787
		(-0.19)		
Composite	Q1	0.939***	11.866	15.564
		(3.23)		
	Q5	0.904***	11.409	16.183
		(2.99)		
	Q5-Q1	−0.034	−0.412	4.865
		(-0.38)		

Notes: Portfolios are formed monthly based on quintiles of drift measures. Returns are equal-weighted. Mean returns are reported in monthly percentage terms. Annualized returns are computed as 12× the arithmetic mean of monthly returns. Volatility is the annualized standard deviation of monthly returns, computed as $\sqrt{12}$ times the standard deviation of monthly returns. t-statistics (Newey-West with 6 lags) are reported in parentheses. ***, **, * denote statistical significance at the 1%, 5% and 10% levels, respectively.

5.2 Factor model alphas

To determine whether the long-short spread documented in Table 3 reflects compensation for exposure to priced risk factors or a genuine anomaly, we regress the monthly excess returns of the long-short portfolio for each signal on four factor models. Table 4, Panel A, reports the intercept (α) from each specification. The abnormal returns associated with the Item 1,1A,7 separately and compounded of the long-short portfolio are consistently negative and grow in economic magnitude as the factor model becomes more stringent. The CAPM (Sharpe, 1964) alpha of −0.359% per month ($t = -2.70$) strengthens to

-0.368% ($t = -2.92$) under the FF5 model (Fama & French, 2015). The monotonic pattern indicates that if the Item 1A anomaly were driven by covariation with FF5 factors, the alpha should shrink rather than grow as these factors are added to the specification. The fact that the FF5 alpha exceeds the CAPM alpha in absolute value indicates that the factor exposures particularly the investment factor CMA (see Panel C) contribute positively to the long-short return and therefore that the unexplained component is larger than the spread. The Fama-French three-factor model (FF3) (Fama & French, 1993) tests whether the long-short returns are attributable to exposures to size and value factors. The alphas remain statistically significant and do not decline in magnitude relative to the CAPM, indicating that neither size nor value risk explains the predictive power of the drift signal. Augmenting the model with momentum (UMD) similarly leaves the results largely unchanged. This is notable given that the portfolios exhibit modest negative loadings on momentum (Panel B), implying a tilt against past winners. Since momentum delivered positive returns over much of the sample, this exposure depresses the raw returns of the strategy rather than inflating them. Controlling for momentum therefore slightly increases the magnitude of the estimated alpha, reinforcing the conclusion that the anomaly is not driven by return continuation effects. Items 1, 7 and the composite produce statistically significant FF5 alphas ($t = -2.69$, -2.91 and -3.03 respectively) even though their raw long-short spreads are insignificant. This can appear paradoxical but it follows directly from the factor structure revealed in Panel B. The Item 7 long-short portfolio loads strongly and negatively on HML ($\hat{\beta}_{HML} = -0.293$, $t = -7.61$), implying that the portfolio is long growth firms and short value firms. Because the value premium was negative during much of our sample period, this exposure depresses the raw return of the portfolio rather than inflating it. Once HML is controlled for, the alpha is revealed as statistically significant. The Item 1 and composite portfolios display similar, though smaller, negative HML loadings. The factor loadings in Panels B and C reveal a second pattern worth noting. The Item 1A long-short portfolio exhibits a large positive loading on the conservative-minus-aggressive investment factor (CMA, $\hat{\beta} = 0.219$, $t = 2.95$ under FF5). This implies that firms with high risk-disclosure drift invest conservatively. This means that firms revising their risk factors may be doing so in anticipation of or in response to shifting investment opportunities and may be curtailing capital expenditures accordingly. The fact that this exposure is positive while the alpha remains negative means the investment factor cannot rationalize the return pattern.

All together, the classic factor models indicate that the predictive power of semantic drift is not captured by standard risk factors. The negative alphas persist. In case of Items 1, 7 and the composite, significance is only revealed after controlling for factors that mask the underlying signal. This is the pattern we expect if semantic drift captures information about future fundamentals that is initially underreacted to by investors, rather than compensation for a known source of systematic risk.

Table 4: Return Predictability and Factor Loadings

	Item 1	**Item 1A**	**Item 7**	**Composite**
Panel A: Long-Short Alphas Across Factor Models				
CAPM	-0.1791*	-0.3590***	-0.1978	-0.2041**
	[-1.94]	[-2.70]	[-1.60]	[-2.14]
FF3	-0.1932**	-0.3275**	-0.2458***	-0.2179**
	[-2.31]	[-2.43]	[-2.58]	[-2.56]
FF3 + Mom	-0.1737**	-0.3179**	-0.2286**	-0.1990**
	[-2.02]	[-2.45]	[-2.41]	[-2.35]
FF5	-0.2250***	-0.3682***	-0.2825***	-0.2525***
	[-2.69]	[-2.92]	[-2.91]	[-3.03]
Panel B: Factor Loadings (FF3 + Momentum)				
α	-0.1737**	-0.3179**	-0.2286**	-0.1990**
	[-2.02]	[-2.45]	[-2.41]	[-2.35]
MKT	0.030	-0.034	0.067	0.036
	(1.34)	(-1.78)	(2.61)	(1.48)
SMB	0.012	0.013	0.012	0.028
	(0.37)	(0.32)	(0.30)	(0.72)
HML	-0.121	0.112	-0.293	-0.134
	(-3.49)	(2.44)	(-7.61)	(-3.37)
UMD	-0.064	-0.029	-0.056	-0.062
	(-3.29)	(-1.18)	(-2.07)	(-2.29)
R^2	0.101	0.056	0.297	0.111
Panel C: Factor Loadings (FF5)				
α	-0.2250***	-0.3682***	-0.2825***	-0.2525***
	[-2.69]	[-2.92]	[-2.91]	[-3.03]
MKT	0.056	-0.005	0.087	0.065
	(2.95)	(-0.28)	(3.55)	(2.63)
SMB	0.046	0.044	0.050	0.062
	(1.34)	(0.99)	(1.39)	(1.71)
HML	-0.126	0.032	-0.284	-0.161
	(-3.34)	(0.65)	(-5.30)	(-2.80)
RMW	0.068	0.033	0.099	0.055
	(1.29)	(0.51)	(1.86)	(0.95)
CMA	0.055	0.219	0.009	0.112
	(0.93)	(2.95)	(0.12)	(1.57)
R^2	0.080	0.086	0.290	0.101

Notes: This table reports long-short portfolio alphas and factor loadings for drift-based strategies. Panel A reports alphas across asset pricing models. Panels B and C report full factor loadings. Alphas are monthly returns multiplied by 100. Factor loadings are in levels. Newey-West (6 lags) t-statistics are reported in brackets or parentheses. Sample sizes are 239, 230, 239 and 239 observations for Item 1, Item 1A, Item 7 and the composite, respectively. ***, **, * denote statistical significance at the 1%, 5% and 10% levels.

5.3 Fama-MacBeth Regressions (Per Signal)

Table 5 reports Fama-MacBeth cross-sectional regressions for each drift signal, with specifications ranging from univariate (U) to fully controlled (C). In the controlled specifications, drift signals enter alongside momentum, short-term reversal, log market capitalization, book-to-market and return on assets.

The results complement the portfolio evidence. Item 1A is the only signal whose univariate Fama-MacBeth coefficient is statistically significant ($\hat{\beta} = -0.0482$, $t = -2.00$), consistent with its role as the strongest predictor of future returns. In the full regression with all controls, the coefficient remains negative and significant under the 10% significance level ($\hat{\beta} = -0.0451$, $t = -1.74$). Item 1 becomes significant under the 10% level with controls ($\hat{\beta} = -0.0366$, $t = -1.89$), suggesting that its predictive content is partially concealed by correlation with firm characteristics in the univariate regression. Item 7 and the composite remain insignificant in both regressions.

Book-to-market enters strongly negative ($\hat{\beta} \approx -2.25$, $t \approx -10$ across specifications), as does ROA ($\hat{\beta} \approx -5.5$, $t \approx -7$). Log market capitalization is strongly positive. This collection of signs is unusual in the classical cross-sectional asset pricing literature, where book-to-market and profitability typically enter with positive signs. Our estimates reflect the specific nature of our sample: S&P 500 firms over 2005-2024, a period characterized by the sustained outperformance of large-cap growth stocks relative to small-cap value stocks. The coefficients should therefore be interpreted as conditional on our sample composition rather than as estimates of unconditional risk premia. Importantly, this observation does not compromise the interpretation of our drift coefficients: the signal's incremental predictive power is identified relative to the sample-specific pricing of firm characteristics.

The average R^2 rises sharply from roughly 0.003 in the univariate specifications to approximately 0.09 when controls are included. The bulk of this increase is attributable to book-to-market, ROA and market capitalization, not the drift signal itself. This is not evidence against the drift signal; rather, it reflects the standard pattern in cross-sectional return regressions where known firm characteristics account for a substantial share of explainable monthly cross-sectional variation, leaving little residual variation for any individual signal to explain. What matters for our purposes is that the Item 1A coefficient retains its sign and approximate magnitude after controlling for these characteristics.

Table 5: Fama-MacBeth Regressions (Per Signal)

	Item 1		Item 1A		Item 7		Composite	
Variable	**U**	**C**	**U**	**C**	**U**	**C**	**U**	**C**
Item 1	−0.0068	-0.0366*						
	(-0.31)	(-1.89)						
Item 1A			-0.0482**	-0.0451*				
			(-2.00)	(-1.74)				
Item 7					−0.0142	−0.0231		
					(-0.60)	(-1.18)		
Composite							−0.0232	−0.0328
							(-1.04)	(-1.64)
Momentum		-0.5512*		-0.6683**		-0.5333*		-0.5350*
		(-1.91)		(-2.22)		(-1.87)		(-1.87)
Reversal		−0.9383		-1.2594*		−0.9395		−0.9331
		(-1.43)		(-1.72)		(-1.44)		(-1.41)
log(MktCap)		0.3159***		0.3370***		0.3154***		0.3165***
		(6.60)		(6.55)		(6.59)		(6.62)
Book/Market		-2.2480***		-2.2496***		-2.2110***		-2.2309***
		(-10.48)		(-9.71)		(-10.11)		(-10.34)
ROA		-5.5138***		-5.4047***		-5.4853***		-5.5142***
		(-6.99)		(-6.41)		(-6.86)		(-6.90)
Intercept	0.9542***	−0.7898	0.9322***	−1.0156	0.9556***	−0.8023	0.9551***	−0.8019
	(3.31)	(-1.25)	(2.92)	(-1.50)	(3.32)	(-1.27)	(3.31)	(-1.27)
Months (T)	239	239	215	215	239	239	239	239
N	463	449	462	447	464	450	468	454
R^2	0.0029	0.0918	0.0032	0.0949	0.0028	0.0907	0.0027	0.0912

Notes: Coefficients are multiplied by 100. t-statistics (Newey-West with 6 lags) are in parentheses. ***, **, * denote significance at the 1%, 5% and 10% levels, respectively.

5.4 Return predictability of different holding periods

To investigate the time span in which semantic drift information is incorporated into prices, we re-estimate the portfolio sorts, Fama-MacBeth regressions and factor-model alphas with return windows of 1, 3, 6 and 12 months after the 10-K filing date. Table 6 reports these results.

First, at short horizons (one and three months) no signal produces a statistically significant long-short spread or Fama-MacBeth coefficient. Second, significance emerges predominantly at the twelve-month horizon across all three metrics the long-short spread, the Fama-MacBeth coefficient and the Carhart alpha. Third, the twelve-month Carhart alphas are statistically significant for Item 1 (-0.184%, $t = -2.14$), Item 7 (-0.252%, $t = -2.66$) and the composite (-0.234%, $t = -2.80$), broadening the pattern of predictability

relative to the raw spread comparison in Table 3.

The absence of a short-term reaction in equity returns is in itself an economical finding. If investors and analysts covering these companies are examining the textual content of 10-K disclosures at the time of filing, one would expect prices to adjust within days (or even seconds in digital times) of publication and leave no predictable return pattern at subsequent periods. The delayed incorporation of the signal is consistent with the behavioral limited-attention framework of Hirshleifer and Teoh (2003) and with the "lazy prices" interpretation of Cohen et al. (2020). In this behavioral view, investors do not process the full informational content of annual reports without incurring costs. Investors instead update their equity valuations gradually as the strategic shifts and risk revisions hinted at in the text manifest in subsequent quarterly earnings announcements and other publicly observable outcomes such as news channels or papers throughout the year.

A potentially counterintuitive result in the horizon analysis is that within the one-month period, Item 1 produces a small positive Carhart alpha of +0.809% ($t = 1.77$). We interpret this as an side effect of limited statistical power with a single-month return window rather than as evidence of genuine short-horizon predictability. This is because the Fama-MacBeth and long-short spread estimates at the one-month horizon for Item 1 are insignificant and of opposite sign, consistent with sampling variability.

Table 6: Return Predictability by Holding-Period Horizon

Signal	**Horizon**	**LS Spread (%)**	**FM β**	**α (Carhart, %)**
Item 1	1m	0.5832 (1.12)	-0.0457 (-0.29)	0.8091* (1.77)
	3m	0.0157 (0.07)	-0.1195 (-1.22)	-0.0064 (-0.03)
	6m	0.1783 (1.02)	-0.0326 (-0.59)	0.1044 (0.51)
	12m	-0.0155 (-0.17)	-0.0342* (-1.73)	-0.1838** (-2.14)
Item 1A	1m	-0.3313 (-0.72)	-0.0041 (-0.02)	-0.2558 (-0.61)
	3m	-0.2273 (-0.77)	0.0228 (0.19)	-0.2440 (-0.88)
	6m	-0.1545 (-0.79)	-0.1232* (-1.85)	-0.2110 (-0.95)
	12m	-0.1880 (-1.64)	-0.0513* (-1.94)	-0.2516* (-1.97)
Item 7	1m	-0.0127 (-0.02)	-0.5379 (-0.99)	-0.2495 (-0.48)
	3m	0.2276 (0.80)	-0.1981 (-0.85)	-0.0026 (-0.01)
	6m	0.1335 (0.87)	-0.1283 (-1.11)	-0.1124 (-0.80)
	12m	-0.0443 (-0.41)	-0.0266 (-1.39)	-0.2523*** (-2.66)
Composite	1m	0.1933 (0.39)	-0.1886 (-0.75)	0.1722 (0.35)
	3m	-0.1895 (-0.69)	-0.0935 (-0.72)	-0.3176 (-1.00)
	6m	-0.0718 (-0.45)	-0.1011 (-1.01)	-0.2372 (-1.29)
	12m	-0.0669 (-0.74)	-0.0378* (-1.91)	-0.2339*** (-2.80)

Notes: This table reports return predictability across different holding-period horizons. LS Spread denotes the return difference between the highest and lowest drift quintiles. FM β denotes the average slope coefficient from Fama and MacBeth (1973) cross-sectional regressions, standardized by one standard deviation of the signal. α is the Carhart four-factor (Carhart, 1997) alpha. All return measures are monthly percentages. t-statistics (Newey-West with 6 lags) are reported in parentheses. ***, **, * denote statistical significance at the 1%, 5% and 10% levels, respectively.

5.5 Interaction effects

Table 7 reports Fama-MacBeth regressions in which each drift signal is interacted with log market capitalization, book-to-market and ROA. These tests assess whether the predictive content of semantic drift is concentrated in firms for which the limited-attention channel is most plausible.

The interaction between drift and book-to-market is negative and statistically significant for every signal we test: Item 1 ($\hat{\beta} = -0.0889$, $t = -3.32$), Item 1A ($\hat{\beta} = -0.0534$, $t = -2.06$), Item 7 ($\hat{\beta} = -0.1634$, $t = -3.04$) and the composite ($\hat{\beta} = -0.1075$, $t = -4.10$). The interpretation is that the negative return-predictability of drift is substantially stronger for firms with higher book-to-market ratios, i.e., value firms. This pattern is consistent with a cognitive or economic channel. For a mature value firm, an abrupt semantic shift in the 10-K likely signals a departure from a stable operating model and raises concerns about execution risk. For a growth firm, by contrast, strategic repositioning is priced into valuations ex ante, so the marginal informational content of a given drift observation is smaller.

The size interaction is statistically significant and positive for Item 1 ($t = 2.32$) and Item 7 ($t = 1.86$). Given that the main effect on drift is negative, a positive interaction with size implies that the drift penalty is larger in absolute value for smaller firms than for larger firms. This is precisely the pattern predicted by limited-attention models: smaller firms have less analyst coverage, less institutional following and less automated machine-readership of their disclosures, so textual signals about them are likely to be underreacted to more severely. For Item 1A, the size interaction is insignificant, which may reflect the relative uniformity of investor attention to risk-factor disclosures across firm sizes within the S&P 500.

The interaction with ROA is less systematic. We find a significant positive interaction for Item 1 ($t = 1.89$), implying that the drift penalty is larger for low-ROA firms and insignificant interactions elsewhere. Economically, this partial evidence suggests that drift is more informative when combined with signals of operating distress, but the result is not robust across sections.

Taken together, the interaction analysis supports two complementary interpretations. The robust negative drift-by-book-to-market interaction is consistent with an economic channel in which the severity of a disclosure change depends on the firm's life-cycle stage. The positive drift-by-size interactions for Items 1 and 7 are consistent with a behavioral channel in which limited attention amplifies the mispricing of disclosure changes for less-covered firms. These channels are not mutually exclusive and the empirical evidence is consistent with both operating simultaneously.

Table 7: Fama-MacBeth Regressions with Interaction Terms

Signal	**Interaction**	β**(Drift)**	β**(Interact)**
Item 1	Size	-0.045** (-2.14)	0.059** (2.32)
	Book/Market	-0.042** (-2.08)	**-0.089*** (-3.32)**
	ROA	-0.087** (-2.56)	0.728* (1.89)
Item 1A	Size	-0.045* (-1.69)	0.004 (0.16)
	Book/Market	-0.041 (-1.51)	**-0.053** (-2.06)**
	ROA	-0.065 (-1.51)	0.443 (0.75)
Item 7	Size	-0.023 (-0.67)	0.066* (1.86)
	Book/Market	-0.044 (-1.39)	**-0.163*** (-3.04)**
	ROA	-0.034 (-0.83)	0.155 (0.26)
Composite	Size	-0.036 (-1.53)	0.031 (1.22)
	Book/Market	-0.022 (-1.01)	**-0.108*** (-4.10)**
	ROA	-0.057 (-1.63)	0.499 (1.04)

Notes: This table reports Fama and MacBeth (1973) regressions with interaction terms between semantic drift and firm characteristics. The reported coefficients correspond to the main effect of drift and its interaction with size (log market capitalization), book-to-market and return on assets (ROA). Coefficients are multiplied by 100. t-statistics based on Newey-West (6 lags) standard errors are reported in parentheses. The sample spans 215-239 monthly cross-sections depending on the specification, with an average of approximately 447-454 firms per month. ***, **, * denote statistical significance at the 1%, 5% and 10% levels, respectively.

In summary, the results in this section point to three main conclusions. First, semantic drift in the Risk Factors section of 10-K filings predicts negative future stock returns and the magnitude is economically meaningful. The estimated effect corresponds to roughly -4.4% annualized alpha under the FF5 model. This is important given that the analysis focuses on S&P 500 firms, where markets are typically assumed to be efficient and mispricing should be limited.

Second, this predictive pattern cannot be explained by standard risk factors. Controlling for market, size, value, profitability, investment and momentum does not eliminate the effect. If anything, the opposite happens: the alphas become slightly stronger as additional factors are introduced. This suggests that the return predictability is not compensation for known sources of systematic risk, but reflects something not captured by conventional asset pricing models.

Third, the way the signal is incorporated into prices is gradual rather than immediate. The return effect unfolds over the twelve months following the filing, which is consistent with a delayed market reaction. Moreover, the effect is not uniform across firms. It is more pronounced among value firms and among smaller constituents within the S&P 500, where investor attention is likely to be more limited. This cross-sectional pattern supports the idea that the mechanism operates through an attention-based channel rather than through risk.

Finally, there is meaningful heterogeneity across sections of the 10-K. The Risk Factors section (Item 1A) contributes most to the results, while Items 1 and 7 are less indicative. Contrary to the existing literature, which aggregates text across the entire document, the findings here suggest that where the information appears matters. In particular, Item 1A emerges as the most informative part of the filing for capturing economically relevant textual signals.

6 Discussion

Our empirical findings provide support for the "lazy prices" hypothesis, demonstrating that the market systematically underreacts to strategic and risk-related textual shifts in corporate disclosures. However, by disentangling the 10-K into sections, our analysis reveals that the predictive power of semantic drift is not uniform. We find that Item 1A (Risk Factors) serves as the strongest predictor of future negative abnormal returns, outperforming Item 1 (Business) and Item 7 (MD&A). This divergence carries strong economic intuition. Because managers are legally incentivized to disclose emerging threats in Item 1A to avoid litigation, material shifts in this section represent genuine, forward-looking operational risks. Furthermore, because the Risk Factors section is historically characterized by dense, repetitive boilerplate language, investors are particularly prone to limited attention here. When management subtly but meaningfully alters the semantic structure of their risk disclosures, the market acts lazily, failing to incorporate this heightened uncertainty into the stock price until the underlying risks eventually materialize.

While our results align with the directional findings of Cohen et al. (2020) (that greater textual modification predicts lower future returns) our methodological framework isolates a fundamentally different mechanism. Cohen et al. (2020) relied on TF-IDF cosine similarity, which measures exact word overlap and frequency changes. In contrast, our deployment of the `all-MiniLM-L6-v2` model captures shifts in deep semantic context. This distinction is important: a firm could rewrite its 10-K using entirely new vocabulary while maintaining the same strategic posture, which would trigger a false positive under Bag-of-Words methods. Our findings suggest that the lazy prices phenomenon extends beyond mere surface-level editorial revisions to actual, substantive repositioning in corporate narrative meaning.

The horizon analysis detailed in Table 6 further illuminates the mechanics of investor inattention. We observe that the return predictability of the semantic drift signal is largely statistically insignificant at the one-, three- and six-month horizons, only demonstrating robust significance at the twelve-month horizon. This delayed reaction provides some evidence for the limited attention channel. Rather than pricing the informational content of the 10-K upon the filing date, the market appears to wait until the strategic pivots or heightened risks hinted at in the text finally manifest in subsequent fundamental outcomes, such as quarterly earnings surprises or cash flow deteriorations throughout the following year.

This behavioral friction is further supported by the interaction effects documented in Table 7. We find a consistently negative and significant interaction between semantic drift and the book-to-market ratio across signals. This indicates that the penalty for

semantic drift is more severe for value firms than for growth firms. Economically, value firms are often mature or distressed entities where sudden, radical strategic shifts signal elevated execution risk, whereas growth investors may be more tolerant of strategic pivots. Additionally, the positive interaction between Item 1 drift and firm size implies that the negative return predictability is most pronounced among the smallest firms in our sample. This aligns with information asymmetry theories, as smaller firms suffer from lower analyst coverage and less institutional machine-readership, allowing negative textual signals to remain unpriced for longer periods.

A concern regarding our interpretation is endogeneity. There is a possibility of reverse causality between semantic drift and subsequent returns. If a firm is already experiencing deteriorating fundamentals when the 10-K is filed, management may rewrite its risk factors and business description to reflect the new situation of the firm. If this is the case, the relationship between semantic drift and returns would not reflect investor inattention to new information, the two would both simply be outcomes of the same underlying problem. The drift and return are then driven by the same underlying bad news rather than drift actually predicting returns. Our design addresses this to some extent by using a one-year lag, but we cannot fully rule out that the drift is just a proxy for bad fundamentals that get priced in slowly. Testing this properly would require an instrumental variable or a natural experiment.

Our choice of factor models also deserves some comment. We test the signal against the CAPM, the Fama-French three-factor model, the Carhart four-factor model and the Fama-French five-factor model. However, we do not test it against the q-factor model of Hou, Xue and Zhang (2015), the mispricing factors of Stambaugh and Yuan (2017), industry-adjusted returns or characteristic-matched benchmarks. The q-factor and mispricing-factor models are specifically built to absorb return patterns tied to investment and profitability, so a signal that survives the five-factor model may not survive these more demanding specifications. Industry-adjusted returns and characteristic-matched benchmarks would address the concern that our signal is just capturing sector-specific or characteristic-specific return patterns that the standard factor models do not fully strip out. Testing our signal against these alternative benchmarks is an interesting next step.

Despite these robust findings, our study faces several limitations that present avenues for future research. First, to ensure a high-quality, bias-controlled sample, we restricted our universe to the S&P 500. These are the most heavily scrutinized firms in the global equity market; finding significant alpha within this universe is notoriously difficult. Expanding the semantic drift methodology to a broader scope, where information frictions and retail investor inefficiencies are more pronounced, would likely yield even stronger return predictability. Second, our portfolio sorts rely on equal-weighted returns, meaning the documented alphas may be partially driven by the smaller constitu-

ents of the S&P 500. Finally, from a computational perspective, the 512-token limit of the `all-MiniLM-L6-v2` model required us to segment the massive 10-K documents into chunks and compute a mean-pooled composite embedding. While effective, mean-pooling risks diluting highly concentrated, localized strategic disclosures. Future research leveraging newer Large Language Models with expanded context windows could capture the holistic semantic structure of corporate filings with even greater precision.

7 Conclusion

In this paper, we demonstrate that the Semantic Drift signal, particularly when captured from Item 1A (Risk Factors) of 10-K filings, generates statistically significant return predictability among S&P 500 firms over the 2005-2025 period. Specifically, a long-short portfolio formed on the Item 1A Semantic Drift score produces a statistically significant Fama-French five-factor monthly alpha of -36.82 basis points. We show that this predictive signal remains robust across alternative asset pricing specifications, including the CAPM, the Fama-French three-factor model and the Carhart four-factor model. Furthermore, the signal's predictive power persists in Fama-MacBeth cross-sectional regressions after controlling for size, book-to-market ratio, profitability, momentum and short-term reversal. While a composite signal derived from Items 1, 1A and 7 combined delivers comparable statistical significance, Items 1 and 7 show considerably weaker return predictability when evaluated in isolation.

Beyond documenting this baseline predictability, our analysis pinpoints where and when the mispricing occurs. The return effect is concentrated at the twelve-month horizon and is statistically insignificant at shorter windows, consistent with a limited-attention channel in which the market prices the informational content of 10-Ks only once the underlying risks materialize in realized fundamentals. The effect is also stronger among value firms and among the smallest constituents of our sample, aligning with information-asymmetry explanations for delayed price discovery. Methodologically, our use of transformer-based embeddings rather than bag-of-words similarity shows that the lazy prices phenomenon extends beyond editorial revisions of Cohen et al. (2020) to substantive shifts in narrative meaning, even when firms rewrite disclosures using entirely new vocabulary.

Our findings are subject to several limitations. Our sample is restricted to the S&P 500, where intense analyst coverage and automated machine-readership likely incorporate informational edges from annual disclosures relatively quickly; extending the framework to broader, smaller-cap universes with greater information frictions could be a next step. Our portfolio sorts are also equal-weighted, so value-weighted designs are needed to confirm the documented alphas are not driven by the smallest constituents of our sample. We further cannot fully rule out reverse causality, since deteriorating fundamentals may prompt the very disclosure changes we measure and our signal has not been tested against the q-factor model of Hou et al. (2015), the mispricing factors of Stambaugh and Yuan (2017) or characteristic-matched benchmarks. Finally, the 512-token limit of `all-MiniLM-L6-v2` required segmenting filings into chunks and computing mean-pooled embeddings; future research leveraging Large Language Models with expanded context windows could capture the whole semantic structure of corporate filings with greater precision.

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